O-S-635D
May 4, 1977
SUPERSEDING
O-S-635C
September 13, 1965

FEDERAL SPECIFICATION

SODIUM POLYPHOSPHATES, TECHNICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers three types and three forms of sodium polyphosphates (see 6.1).

1.2 Classification. Sodium polyphosphates shall be of the following types and forms, as specified (see 6.2):

1.2.1 Types:

Type I - Sodium tetraphosphate.
Type II - Sodium hexametaphosphate.
Type III - Sodium tripolyphosphate.

1.2.2 Forms:

Form 1 - Powder.
Form 2 - Granular (crushed).
Form 3 - Plate.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.
UU-S-48 - Sacks, Shipping, Paper.
PPP-B-35 - Bags, Textile, Shipping, Burlap, Cotton and Waterproof Laminated.
PPP-D-723 - Drums, Fiber.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Std. No. 313 - Material Safety Data Sheets, Preparation and Submission Of.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 501 - Standard Methods of Sampling and Chemical Analysis of Alkaline Detergents.
D 502 - Standard Method of Test for Particle Size of Soaps and Other Detergents.
E 11 - Standard Specification for Wire-Cloth Sieves for Testing Purposes.
E 70 - Standard Method of Test for pH of Aqueous Solutions with the Glass Electrode.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Particle size.

3.1.1 Forms 1 and 2. Forms 1 and 2 shall conform to the particle size as specified in table I when tested as specified in 4.3.1.1.

TABLE I. Particle sizes of form 1 and form 2

Requirement	Form 1	Form 2
Percent retained on:		
3.36 mm (No. 6) sieve	-	10 (max)
600 um (No. 30) sieve	5 (max)	-
150 um (No. 100) sieve	40 (max)	80 (min)

3.1.2 Form 3. Form 3 may be irregularly-shaped plates and shall measure not greater than 4 in. (10.2 cm) on an edge and 1/4 in. (0.6 cm) in thickness when tested as specified in 4.3.1.2.

3.2 Chemical requirements. Sodium polyphosphate shall conform to the requirements of table II when tested as specified in 4.3.2 through 4.3.4.

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TABLE II. Chemical requirements for sodium polyphosphates

Requirement	Type I	Type II	Type III
Total P_2O_5, weight percent, minimum	62.8	66.5	56.0
pH, 1 percent solution	7.2 - 8.3	6.6 - 7.3	9.3 - 10.4
Matter insoluble in water, weight percent, maximum	0.1	0.1	0.1

3.3 Workmanship. Sodium polyphosphate shall be visibly free from foreign matter and suitable for its intended use when examined as specified in 4.2.3.

3.4 Safety data. Safety data shall be submitted in accordance with the Fed. Std. No. 313.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Examination of preparation for delivery.

4.2.1.1 Lot. For this examination, a lot shall be sodium polyphosphate fully packed, marked, and offered for examination at one time. Lot size shall be expressed in number of filled shipping containers.

4.2.1.2 Sampling plan. Sample unit for this examination shall be one shipping container. Sampling shall be conducted in accordance with MIL-STD-105. Sample size, acceptance and rejection numbers shall be determined by special inspection level S-2 and acceptable quality level (AQL) 4.0 percent defective. Defects shall be scored in accordance with table III.

TABLE III. Examination of preparation for delivery

Examine	Defect
Marking	Missing, incorrect, illegible; not as specified.
Packing	Not as specified.
Contents	Net contents less than specified.
Workmanship	Container bulged, distorted, or otherwise defective; tear, cut, or hole; evidence of leakage from any part of the container.

4.2.2 Examination of closure. When the shipping container is required to comply with UU-S-48 or PPP-B-35, examination for defects in closure shall be in accordance with the appendix of the appropriate specification.

4.2.3 Examination of workmanship.

4.2.3.1 Lot. For this examination, a lot shall be one type and form of sodium polyphosphate produced by one manufacturer, at one plant, from the same raw materials, by the same manufacturing process, and offered for examination at one time. Lot size shall be expressed in number of pounds of sodium polyphosphate.

4.2.3.2 Sampling plan. Sample size for this examination shall be one pound (approx. 500 g) drawn at random from the lot. Lot shall be rejected if the sample contains foreign matter or is not suitable for its intended use.

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4.2.4 Testing.

4.2.4.1 Lot. A lot shall be as specified in 4.2.3.1.

4.2.4.2 Sampling plan. Sampling shall be as specified in 4.2.3.2. Sample shall be sealed in a clean, dry, moisture-proof container with identifying lot number and date. Sample shall be tested as specified herein. Lot shall be rejected if sample fails one test.

4.3 Test methods.

4.3.1 Particle size.

4.3.1.1 Forms 1 and 2. Particle size of forms 1 and 2 shall be determined in accordance with ASTM D 502, except that sampling shall be as specified in 4.2.3.2 and sieves shall be as specified in 3.1.1. Sieves shall conform to ASTM E 11. Percentage of sodium polyphosphate retained on the sieves shall be calculated by dividing the weight of the residue on the sieve by the original weight of the sample and multiplying by 100.

4.3.1.2 Form 3. Particle size of form 3 may be visually estimated.

4.3.2 Phosphorus. Total phosphorus, as P_2O_5, shall be determined in accordance with ASTM D 501 for sodium triphosphate, total P_2O_5 by pH titration.

4.3.3 pH. pH of a one percent solution shall be determined in accordance with ASTM E 70.

4.3.4 Insolubles. Insoluble matter shall be determined in accordance with ASTM D 501 for sodium triphosphate, matter insoluble in water.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Fifty (50) or one hundred (100) pounds as specified (see 6.2) of one type and form of sodium polyphosphate shall be packed in a drum conforming to PPP-D-723, type II, grade A. The interior of the drum shall be lined with a polyolefin film of 0.002-in. nominal thickness conforming to L-P-378, type I, class 1, grade A, finish 1. Alternatively, the specified quantity, type, and form of sodium polyphosphate shall be packed in a bag conforming to PPP-B-35, type III, bag No. P3, P6, or P15, liner A or B, or conforming to UU-S-48, level A packing for sodium phosphate.

5.1.2 Commercial. Sodium polyphosphate shall be packed in containers to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.2 Marking. Shipping containers and unitized loads (when applicable) shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as specified (see 6.2). In addition, all containers shall bear the following special marking:

NOT FOR USE IN POTABLE WATER

5.3 Unitization. (Applicable to full rail car and truck load shipments only.) The containers shall be unitized for shipment in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length and 45 inches in width.

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6. NOTES

6.1 Intended use. Sodium polyphosphates covered by this specification are intended for use in industrial water treatment but are not intended for use in water for human consumption. They form soluble complexes with calcium and magnesium and may be used to soften water and control scale formation in condensers, heat exchangers, pipelines, and boilers. Sodium polyphosphates are interchangeable for the intended use, but possess the differences indicated below. Selection of the type and form should depend upon established patterns of usage, compatibility with equipment, and economic advantage.

6.1.1 Type I. Sodium tetraphosphate is primarily intended for use in water treatment for laundries.

6.1.2 Type II. Sodium hexametaphosphate is primarily intended for use in the treatment of boiler feedwater. This type is commonly used for the threshold treatment of industrial water. In those cases where the use of this type requires the addition of a relatively large quantity of caustic soda or soda ash, type III might be preferred.

6.1.3 Type III. Sodium tripolyphosphate is intended for use as a substitute for types I or II whenever it is determined to be necessary for performance or economically advantageous.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and form required (see 1.2).
(c) Whether level A or commercial packing is required (see 5.1).
(d) Unit quantity required (see 5.1.1).
(e) Marking document required (see 5.2).

6.3 Classification changes. This specification supersedes O-S-635C, dated September 13, 1965, and Amendment-1, dated September 9, 1966. The following classification changes have been incorporated as indicated in table IV.

TABLE IV. Classification changes in O-S-635D

Form	O-S-635C	O-S-635D
Powder	Powder	Form 1
Granular	Granular	Form 2
Bead	Bead	(deleted)
Flake	Flake	(deleted)
Plate	Plate	Form 3

MILITARY CUSTODIANS:

Army - EA
Navy - YD
Air Force - 68

Review activities:

Army - EA, MD, SM
Navy - YD
Air Force - 68
DSA-GS

User activity:

Navy - SH

Military coordinating activity:

Army - EA

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
HEW-NIH
INT-MIN
VA-DMS

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price .30 cents each.

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